

"Is This Safe?" — Volunteer Answer Key

FOOD SAFETY

FS-01 SAFE

Sarah stores freshly cooked potato salad in the refrigerator within one hour of cooking and consumes it within 3 days.

Why: Cooling cooked potato salad quickly and refrigerating within 2 hours helps prevent bacterial growth. Consuming within 3-4 days reduces risk of spoilage and pathogens.

Volunteer notes: Following the 2 hour rule for perishable foods and refrigeration at below 40°F prevents bacterial growth. Leftovers should be eaten within 3-4 days for safety. Follow storage time guidelines to avoid foodborne illness.

Source: SP 50-1000 Turkey basics - "Adhere to the 2 hour rule: discard perishables left at room temperature more than 2 hours. Use leftovers within 3 to 4 days so mold doesn't have a chance to grow."

FS-02 SAFE

James decides to wash fresh apples under running water before eating them without peeling.

Why: Washing fresh fruits under running water removes dirt and potential surface microbes, reducing the risk of contamination from handling or environmental sources.

Volunteer notes: Thorough rinsing of fresh fruits and vegetables before eating is a basic food safety step to reduce bacteria and pesticide residues. It lowers risk of foodborne illness.

Source: SP 50-1000 Turkey basics - "Thoroughly rinse fresh fruits and vegetables under running water before eating."

FS-03 NOT SAFE

Lisa stores homemade garlic in oil at room temperature in a jar on her kitchen counter to use over the coming month.

Why: Garlic in oil at room temperature creates an anaerobic environment favorable for *Clostridium botulinum* growth and toxin formation; it must be refrigerated and used quickly.

Volunteer notes: Home-made garlic in oil must be refrigerated and used within 3 weeks to prevent botulism. Commercial garlic in oil is acidified for safety. Room temperature storage risks *C. botulinum* growth.

Source: SP 50-1000 Turkey basics - "Long term refrigerated storage of garlic or herbs in oil is associated with a risk of botulism. Keep homemade garlic in oil refrigerated; use within 3 weeks."

FS-04 NOT SAFE

Tom leaves a cooked stew on the counter for 5 hours before refrigerating it.

Why: Leaving cooked food in the temperature danger zone (40-140°F) for more than 2 hours allows rapid bacterial growth and increased foodborne illness risk.

Volunteer notes: The danger zone is 40-140°F where bacteria grow rapidly, especially between 60-125°F. Food held longer than 2 hours there becomes unsafe and must be discarded to avoid illness.

Source: SP 50-1000 Turkey basics - "Temperature is very important. If the product has been in the danger zone (40-140°F) for longer than 2-3 hours, it could be a safety concern."

FS-05 NOT SAFE

Alex prepares fermented cucumber pickles but skips sanitizing the jar and uses cucumbers with soft spots.

Why: Poor sanitation and starting with spoiled vegetables increase risk of growth of harmful pathogens like *Clostridium botulinum* during fermentation.

Volunteer notes: Safe fermentation requires using fresh produce and thoroughly cleaning and sanitizing containers to encourage beneficial microbes and prevent pathogens and spoilage bacteria.

Source: SP 50-1000 Turkey basics - "Proper sanitation includes thorough washing of hands prior, washing and sanitizing all equipment, and starting with fresh fruits and vegetables."

FS-06 IT DEPENDS

Maria left a vacuum-sealed homemade dried jerky package unrefrigerated during a 3-day camping trip in warm weather.

Why: Safety depends on the drying process and final moisture: properly dried jerky and vacuum-sealed can be room temperature stable, but incomplete drying or warm temps risk bacterial growth.

Volunteer notes: Jerky must be dried to safe moisture level and acid or salt content, then stored properly. Warm temperatures and insufficient drying increase risk of pathogens even in vacuum packaging.

Source: SP 50-1000 Turkey basics - "Properly dried foods with low moisture and salt content can be stored at room temperature safely."

FS-07 IT DEPENDS

John plans to can tomatoes at 5,000 feet elevation using a boiling water bath canner without adjusting processing time.

Why: At altitudes over 1,000 feet, water boils at lower temps, so processing time or pressure must be adjusted for safe acidity and pathogen control in home canning.

Volunteer notes: Altitude affects boiling temperature. At 5,000 feet, longer times or pressure canners are needed to ensure food safety due to lower boiling temps failing to destroy *Clostridium botulinum* spores.

Source: Troubleshooting canning acid foods - "Adjust processing times or pressure for altitudes above 1,000 feet to ensure safe home canning."

FS-08 IT DEPENDS

Emily prepares homemade salad dressing with fresh lemon juice and stores it in the refrigerator. She wonders if it is safe to keep it for 5 days before consuming it.

Why: The safety of homemade salad dressings depends on their acidity and refrigeration. If the dressing is sufficiently acidic (pH 4.6 or below) due to lemon juice and other acidic ingredients, and it is kept refrigerated, it can be safe for several days. However, if the acidity is borderline or other ingredients reduce acidity, bacteria may grow, making it unsafe to keep for 5 days. Always consider the pH, ingredients, and temperature.

Volunteer notes: According to OSU Extension, acidity is key to food safety in preserved and refrigerated foods. Foods with pH above 4.6 are more prone to harmful bacterial growth. Refrigeration slows growth but does not stop it completely. For salad dressings, acidity level and storage temperature are critical. Follow up questions might include: 'How to test acidity of a homemade dressing?' and 'What signs indicate that a dressing went bad despite refrigeration?' Comparative pH charts and refrigeration guidelines assist in answering these.

Source: SP 50-834 Tips for critical thinking - "If the food is low in acid (pH 4.6 and above), we are more concerned about safety than when the food is high in acid (4.6 and below)... Acidic foods include all fruit, tomatoes, and pickles made with an adequate amount of vinegar... Temperature is very important... If the product has been in the danger zone (40 - 140 ° F) for longer than 2 - 3 hours, it could be a safety concern."

CANNING FRUITS

CF-01 SAFE

Julie selects fresh, firm peaches and washes them thoroughly. She peels, slices, and packs the peaches into hot sterilized jars, then covers them with a hot light syrup. She processes the jars in a boiling water canner for the recommended time at her altitude.

Why: Using fresh, firm fruit that is carefully washed and peeled reduces spoilage risks. Hot packing and processing peaches in a boiling water canner for the correct time ensures destruction of harmful microorganisms. Adjusting processing time for altitude is important for safety and quality.

Volunteer notes: According to PNW 199 Canning fruits, fruits can safely be processed in boiling water canners if following tested recipes and adjusting processing times for altitude. Hot packs improve safety by removing air and allowing better heat penetration. Using sterilized jars and proper syrup ensures good preservation. Questions: Why is altitude adjustment needed? Answer: Because water boils at lower temperatures at higher altitudes, processing times must increase to ensure proper heat penetration. What should be done if jars don't seal? Answer: Refrigerate and consume quickly or reprocess with new lids within 24 hours.

Source: PNW 199 Canning fruits - "Fruits ... can safely use a boiling water canner ... processing time adjustment for altitude ... hot syrup and sterilized jars recommended."

CF-02 SAFE

Mark prepares cherries by washing and pitting them. He packs them raw into clean, sterilized jars, fills with boiling water, leaving half-inch headspace, and processes them in a boiling water canner following tested recommendations.

Why: Filling jars with boiling water and packing raw cherries following tested recipes ensures safety. Proper headspace and processing in boiling water canners destroy molds and yeasts and create a vacuum seal that prevents spoilage.

Volunteer notes: PNW 199 recommends raw or hot pack methods with proper headspace and boiling water canning for high-acid fruits like cherries. Headspace allows for expansion and vacuum seal formation. Questions: What is the risk of improper headspace? Answer: Too little headspace can cause overflow; too much can lead to poor seals. Why should

lids not be reused? Answer: Old lids can fail to seal, risking contamination.

Source: PNW 199 Canning fruits - "Raw pack or hot pack methods ... use of boiling water canner ... half-inch headspace ... tested recipes are essential."

CF-03 NOT SAFE

Dana picks wild plums and decides to can them using a pressure canner because the plums are not very acidic. She uses a test from an unverified internet recipe without adding acid.

Why: Plums are borderline acidity fruits and often require added acid for safe boiling water canning; otherwise, a pressure canner is required. Using unverified recipes and omitting acid addition risks survival of botulism-causing bacteria.

Volunteer notes: Low or borderline acid fruits must be processed using pressure canning or acidified according to research-tested recipes to prevent botulism (PNW 199). Unclear recipes are dangerous. Questions: Can you can plums safely without acid? Answer: Only if processed in a pressure canner following tested times. What happens if unsafe recipe is used? Answer: Risk of botulism and spoilage.

Source: PNW 199 Canning fruits - "Some fruits such as plums require acid addition or pressure canning to ensure safety."

CF-04 NOT SAFE

Tom jars fresh strawberries using a 'raw pack' method but does not heat the jars before filling. He leaves them at room temperature before processing 6 hours later in a boiling water canner.

Why: Filling jars with raw fruit into jars that are not hot can cause jar breakage and poor seals. Additionally, waiting many hours before processing allows spoilage organisms to multiply, increasing risk of unsafe product.

Volunteer notes: PNW 199 recommends sterilizing jars by heating and filling jars while hot for fruit. Delays before processing increase spoilage and risk. Questions: Can raw packed fruit be held before processing? Answer: If less than 30 minutes and kept cold. What if jars are cold? Answer: Risk of cracking and seal failure.

Source: PNW 199 Canning fruits - "Jars should be hot when filled to prevent breakage ... work quickly to reduce spoilage."

CF-05 NOT SAFE

Eric relies on the jar lids to seal fruit jam without adjusting processing time for his high-altitude location, which is 4,500 feet above sea level. He processes jars at sea-level time.

Why: Processing times must be increased with altitude because boiling water is at a lower temperature at higher elevations. Using sea-level times at high altitude risks underprocessing, which can cause spoilage or pathogen survival.

Volunteer notes: PNW 199 states processing times must be adjusted for altitude; failing to do so may produce unsafe food. Questions: How much should processing time increase? Answer: Usually add 1 to 5 additional minutes depending on altitude. Can improper processing be fixed? Answer: Reprocess immediately with correct time or discard.

Source: PNW 199 Canning fruits - "Add 1 minute for each 1,000 feet of elevation above 1,000 feet for boiling water canning."

CF-06 IT DEPENDS

Laura decides to can peaches using a homemade syrup recipe with no added sugar because she wants a low-sugar option. She processes the jars in a boiling water bath using standard processing times.

Why: The safety of canning peaches with no sugar depends on whether the syrup provides enough acidity and correct processing time is adjusted. Sugar is not a safety factor but sugar or acidic syrups can help preserve quality. Using tested recipes for low or no sugar syrups and following processing times is important.

Volunteer notes: PNW 199 notes processing times can be the same for unsweetened fruit packed in water as for those packed in syrup, but quality may decline. Ensuring correct headspace and processing time is essential. Questions: Does sugar preserve safety? Answer: No, sugar mainly affects quality and texture. Can syrup be omitted? Answer: Yes, if following a tested recipe and processing time.

Source: PNW 199 Canning fruits - "Sugar is not needed for safety but influences quality; processing times remain the same with water or syrup."

CF-07 IT DEPENDS

Jessica picks Asian pears to can and decides to omit adding lemon juice, believing pears are acidic enough. She uses recipes for boiling water canning without modification.

Why: Asian pears have borderline acidity and typically require acid addition (like lemon juice) to safely process in boiling water canners. Safety depends on the variety's acidity and following a tested recipe. Without acidification, using a pressure canner might be necessary.

Volunteer notes: SP 50-694 notes that Asian pears require acidification for boiling water canning. Using unacidified pears in boiling water canners risks botulism. Questions: How to test acidity? Answer: pH testing or following extension tested recipes. Can pears be pressure canned? Answer: Usually not needed if acidified correctly.

Source: SP 50-694 Preserving asian pears - "Asian pears require acid addition to be safely processed in boiling water canners."

CF-08 IT DEPENDS

Megan plans to can her home-harvested Asian pears using a boiling water canner. She washes, peels, and halves the pears but forgets to add lemon juice to the jars before processing. She uses the raw pack method with syrup and processes the jars for the recommended time at her altitude.

Why: Asian pears are lower in acid than other pear varieties, so USDA and OSU guidelines recommend adding lemon juice to each jar to ensure safe acidity levels during boiling water canning. Without added acid, the safety depends on whether the lemon juice or other acidifying step is included. If lemon juice is omitted, the product may not be safe because pathogenic bacteria like *Clostridium botulinum* could survive. Therefore, safety hinges on whether the acid is added before processing.

Volunteer notes: According to OSU Extension PNW 199 Canning fruits, the recommended procedure for canning Asian pears safely in a boiling water bath includes adding 2 tablespoons bottled lemon juice per quart or 1 tablespoon per pint to the jars before processing. This is necessary because Asian pears have a lower natural acidity, and without acid addition, the product may be unsafe. Volunteers should explain the need for acid addition to prevent bacterial growth, especially botulism. A follow-up question is: 'Can I use fresh lemon juice instead of bottled?' Answer: Bottled lemon juice is preferred since fresh lemon juice acidity varies. Another question: 'What if I use a pressure canner for Asian pears without acid?' Answer: Pressure canning does not guarantee safety for low-acid fruits without acidification, so acid addition is still necessary.

Source: PNW 199 Canning fruits - "Asian pears are lower in acid than Bartlett or other similar varieties of pears. In order to can safely, lemon juice is added to jars before processing. Add 2 tablespoons bottled lemon juice per quart or 1 tablespoon per pint to the jars."

CANNING TOMATOES & SALSAS

CT-01 SAFE

Maria canned a batch of salsa using a tested recipe that includes 5% vinegar, and she processed the jars in a boiling water canner for 20 minutes at her altitude of 2,000 feet.

Why: Using a tested salsa recipe with the right acidity level and processing time in a boiling water canner at the correct altitude prevents the growth of harmful bacteria, including botulism. Vinegar at 5% acidity and following time and temperature guidelines ensure safety.

Volunteer notes: Salsas must be prepared with tested recipes that ensure adequate acidity ($\text{pH} \leq 4.6$) and processed in a boiling water canner for the recommended time, adjusted for altitude, to prevent botulism. Follow exact acid amounts and processing times for safety. Follow-up: What happens if vinegar acidity is lower? (Unsafe, risk of botulism). Can you reduce processing time? (No, must follow tested times).

Source: PNW 395 Salsa recipes for canning - "Salsa recipes for canning with a boiling water canner must meet acidity-level requirements... processed in a boiling water canner 20 minutes at 1,001-6,000 feet altitude"

CT-02 SAFE

John uses Roma tomatoes, peels and cores them, heats salsa ingredients to a boil, and fills hot jars, leaving 1/2 inch headspace before processing in a boiling water canner for 15 minutes at sea level.

Why: Using high-quality tomatoes, peeling and coring them properly, heating salsa to boiling, and processing jars with the correct headspace and time in a boiling water canner meets safety standards. These steps help destroy harmful bacteria and ensure acidity is sufficient.

Volunteer notes: Use only high-quality tomatoes and follow the recipe's preparation instructions (peeling, coring, cooking time). Maintaining headspace and processing time in a boiling water canner is critical for destroying microorganisms. Follow-up: Why peel tomatoes? (To improve texture and remove surface bacteria). Is headspace important? (Yes, to allow heat circulation and seal integrity).

Source: PNW 395 Salsa recipes for canning - "Use high-quality tomatoes... peel by boiling water dip... simmer salsa and fill hot jars leaving 1/2 inch headspace... process in boiling water canner for 15 minutes at 0-1,000 feet"

CT-03 NOT SAFE

Sarah canned fresh tomatoes without adding any lemon juice or citric acid, assuming tomatoes are naturally acidic enough.

Why: Tomatoes must be acidified by adding lemon juice or citric acid before canning to ensure a safe acidity level that prevents botulism. Relying on natural acidity alone is unsafe as tomato pH can vary and sometimes be too high (less acidic).

Volunteer notes: Tomatoes need acidification for safe water bath canning; USDA recommends adding bottled lemon juice or citric acid. Without acidifying, harmful bacteria spores can survive. Follow-up: Can you use fresh lemon juice? (No, use bottled lemon juice or citric acid). What if you forget acid? (Unsafe, possible botulism risk).

Source: PNW 300 Canning tomatoes and tomato products - "Tomatoes with added acid in the form of lemon juice or citric acid... can be canned safely in a boiling water canner... overripe tomatoes or tomatoes from frost-killed vines should not be used because they lose acidity"

CT-04 NOT SAFE

Mike canned a homemade salsa recipe but decided to double the fresh garlic and add fresh herbs before processing.

Why: Increasing fresh garlic or herbs before canning alters the recipe's acidity, potentially lowering it and making the salsa unsafe. Only use tested recipes without altering acid ingredients, as botulism risk depends on acidity level.

Volunteer notes: Fresh garlic and herbs can decrease acidity and increase risk of botulism in salsa. Only use tested recipes and do not add or increase fresh herbs or garlic before canning. Follow-up: Can dried spices be increased? (Yes, flavor only). Can you add after opening? (Yes, safe to add after).

Source: PNW 395 Salsa recipes for canning - "It is not safe to add or increase the amounts of fresh herbs or garlic before canning because they affect the acidity level"

CT-05 NOT SAFE

Lucy packed sliced fresh tomatoes raw into jars and processed them in a boiling water canner without adding any acid or pre-cooking the tomatoes.

Why: Raw-packed tomatoes without added acid do not reach safe acidity levels and may not process to eliminate bacteria. Tomatoes should be acidified and preferably hot-packed (heated) before water bath canning to ensure safety.

Volunteer notes: Raw packing tomatoes without acidification is unsafe. Tomatoes need added acid and heating before processing in boiling water canner. Follow tested recipes for packing method. Follow-up: What is hot pack? (Heating tomatoes before packing jars). Why acidify? (To prevent botulism).

Source: PNW 300 Canning tomatoes and tomato products - "Tomatoes must be acidified with bottled lemon juice or citric acid before canning. Hot pack is recommended for better quality and safety. Raw pack without acid addition is unsafe for water bath canning"

CT-06 IT DEPENDS

David canned a salsa recipe that called for vinegar, but he substituted lime juice instead, using the same amount.

Why: Lime juice can substitute bottled lemon juice in some recipes, but not always vinegar. Vinegar and citrus juices differ in acidity. If the original recipe calls for vinegar, substituting lime juice may alter acidity and safety unless confirmed by research. The safety depends on recipe testing and correct acid levels.

Volunteer notes: Vinegar (5% acidity) can be substituted with bottled lemon or lime juice if recipe calls for vinegar. If lime or lemon juice is called for, vinegar is not a safe substitute. Follow tested recipes strictly. Follow-up: Is all lime juice equal? (No, key lime juice is not safe). When is substitution allowed? (Only equal acid sources with proper pH).

Source: PNW 395 Salsa recipes for canning - "An equal amount of bottled lemon or lime juice can be substituted for vinegar when the recipe calls for vinegar. The reverse is not true: vinegar cannot substitute lemon or lime juice because it is less acidic"

CT-07 IT DEPENDS

Anna canned tomatoes with lemon juice added and processed the jars in boiling water canner at sea level, but she forgot to adjust the processing time for her altitude of 5,500 feet.

Why: Altitude affects boiling temperature and thus processing time. At higher altitudes, longer processing times are needed to ensure safety. If Anna's process time at sea level is too short for 5,500 feet, the salsa might not be safe unless time is appropriately increased.

Volunteer notes: Processing times must be adjusted longer at altitudes above 1,000 feet during boiling water canning. Failure to adjust increases risk of understating and bacterial survival. Follow altitude time charts in recipes. Follow-up: How much extra time? (Usually 5-10 min more). What if not adjusted? (Could lead to unsafe product).

Source: PNW 395 Salsa recipes for canning - "Process boiling water canner times at 0-1,000 feet altitude for 15 minutes; at 1,001-6,000 feet for 20 minutes; above 6,000 feet 25 minutes to ensure safety"

CT-08 IT DEPENDS

Tom packed a mixed vegetable salsa that included carrots, corn, and tomatoes and used a boiling water canner method following his tomato salsa recipe.

Why: Because some vegetables are low-acid foods, including non-acidified vegetables in salsa requires pressure canning for safety, not boiling water canning. The safety depends on the recipe being tested specifically for mixed vegetable salsa and using the correct canning method.

Volunteer notes: Low-acid vegetables like corn and carrots in salsa require pressure canning to destroy bacteria spores. Boiling water canning is safe only for acidified tomato-based salsas. Follow research-tested recipes and correct equipment use. Follow-up: Can you water bath can mixed vegetable salsa? (Only if recipe is tested and acidified). What about pressure canning? (Required for low-acid mixtures).

Source: PNW 172 Canning vegetables - "Vegetables except tomatoes are low-acid and must be pressure canned. Boiling water canner is insufficient for vegetables or vegetable mixtures to prevent botulism"

LOW ACID FOODS

LA-01 SAFE

Jane pressure cans a batch of green beans following an OSU-tested recipe exactly, using a weighted pressure canner at 10 pounds pressure for 25 minutes at 1,000 feet elevation.

Why: Green beans are low-acid foods and must be processed in a pressure canner to prevent botulism. Following tested recipes with correct pressure, time, and altitude adjustments ensures the product is safe.

Volunteer notes: Low acid foods require pressure canning at 240°F (10 lbs weight pressure) to kill *Clostridium botulinum* spores. Following tested procedures for time, pressure, and altitude is critical for safety. Common question: Can water bath canning be used? No, it doesn't reach sufficient temperature for low acid foods.

Source: PNW 361 Canning meat poultry game - "Poultry, red meats, and meat from game animals ... are low-acid foods and must be processed in a pressure canner ... the pressure canner is required to reach the high temperature (240°F) to kill botulism spores."

LA-02 SAFE

Mark freezes fresh corn on the cob for long-term storage by blanching it in boiling water for 4 minutes, cooling it immediately in ice water, then packing it in freezer bags.

Why: Freezing corn after proper blanching is a safe preservation method. Blanching stops enzyme action, and freezing halts microbial growth without safety risks typical to canning low-acid foods.

Volunteer notes: Freezing stops bacterial growth but does not kill spores. Blanching before freezing preserves quality. Unlike canning, freezing doesn't require pressure processing for low acid foods. Question: Is freezing safer than canning? For fresh vegetables, yes, if blanching is done properly.

Source: SP 50-443 Preserving corn - "Blanching vegetables prior to freezing destroys enzymes and improves color and flavor. Freezing halts microbial growth but does not kill bacteria."

LA-03 NOT SAFE

Lara attempts to water bath can fresh green beans in quart jars using boiling water processing for 20 minutes at sea level.

Why: Green beans are low-acid foods that require pressure canning to destroy *Clostridium botulinum* spores. Water bath canning does not reach the temperatures needed for safety, making this process unsafe.

Volunteer notes: Low-acid foods (pH above 4.6) like green beans must be pressure canned. Water bath canning (212°F) is insufficient because botulism spores need 240°F for destruction. Question: Can green beans be safely canned without pressure? No, pressure canning is essential.

Source: PNW 361 Canning meat poultry game - "Because meat and poultry products are low-acid foods, ... Only pressure canning produces temperatures high enough (240°F) to kill ... *Clostridium botulinum*."

LA-04 NOT SAFE

Tom cans homemade salsa combining tomatoes, onions, and peppers following a family recipe without measuring vinegar or processing time.

Why: Salsa is a mix of acid and low-acid ingredients. Without using a tested recipe ensuring correct acidity and processing, botulism risk is high. Vinegar amount and processing methods must be precise.

Volunteer notes: Acidified foods like salsa require tested recipes with proper acid levels to safely process in a boiling

water canner. Unmeasured or altered recipes may lack sufficient acidity, risking botulism. Question: Is adding extra vinegar safer? Not necessarily; the recipe balance is critical.

Source: PNW 395 Salsa recipes for canning - "Because salsas are a mixture of acid and low-acid ingredients ... To avoid this serious form of foodborne illness, follow tested recipes carefully."

LA-05 NOT SAFE

Steve stores freshly minced garlic in olive oil at room temperature for several days before refrigeration.

Why: Garlic in oil is low-acid and an anaerobic environment where botulism spores can grow. Without acidification or refrigeration, this storage method is unsafe and can cause botulism.

Volunteer notes: Homemade garlic-in-oil blends pose botulism risks unless made with acidification and proper refrigeration. Botulism bacteria grow in low-acid, oxygen-free environments at room temperature. Question: Can oil be used for preservation? Only with acid and refrigeration.

Source: SP 50-701 Herbs and vegetables in oil - "Low acid cooked foods left at room temperature, ... homemade mixtures of raw garlic stored in oil at room temperature have caused outbreaks."

LA-06 IT DEPENDS

Anna pressure cans cooked beef stew in pint jars using a dial-gauge pressure canner set at 10 pounds pressure, but she is unsure if her altitude adjustment is correct.

Why: Pressure and processing time must be adjusted for altitude. If Anna's altitude adjustment is incorrect, the internal temperature may not reach levels needed to kill botulism spores, making the stew unsafe. Correct altitude adjustments ensure a safe product.

Volunteer notes: Pressure canning lid pressure must be increased with altitude: 10 lbs at sea level, higher at elevations above 1,000 or 2,000 feet depending on gauge type. Incorrect pressure causes underprocessing. Question: How to find correct pressure? OSU/USDA charts provide this, or have gauge tested annually.

Source: PNW 361 Canning meat poultry game - "Increase pressure at altitudes above 1,000 feet for weighted-gauge canners or 2,000 feet for dial-gauge canners to reach 240°F for pressure canning."

LA-07 IT DEPENDS

Ella ferments green beans in brine following a tested recipe but decides to store them in an airtight container indoors without monitoring temperature.

Why: Proper fermentation produces acid that prevents botulism. However, storing fermented low-acid vegetables in an airtight container without temperature control can be risky if acid production was insufficient or temperature is too warm, allowing harmful bacteria growth.

Volunteer notes: Fermentation safety depends on achieving and maintaining sufficient acidity (pH below 4.6). Airtight containers may trap gases and create anaerobic conditions for botulism if acid levels are inadequate. Question: How to ensure safety? Use tested recipes, monitor pH, and store cool.

Source: PNW 450 Canning smoked fish at home - "Clostridium botulinum ... grow in oxygen-free conditions in low acid foods (pH above 4.6). The production of sufficient acid inhibits growth."

PICKLING

PK-01 SAFE

Susan followed a tested quick-pickle recipe exactly, using white distilled vinegar with 5% acidity, fresh dill, and soaked cucumbers in ice water before packing. She processed the jars immediately and stored them in a cool place.

Why: Susan used a tested recipe with proper vinegar acidity (5%), which is crucial for safety in quick pickles. Using fresh ingredients and soaking cucumbers in ice water helps keep pickles firm and safe. Immediate processing and proper storage further ensure safety.

Volunteer notes: According to PNW 355 Pickling vegetables, using vinegar with 5% acidity is essential for safety; altering vinegar or water ratio is unsafe. Soaking cucumbers in ice water is a safe method for crispness. Tested recipes and proper processing are critical to avoid foodborne illness.

Source: PNW 355 Pickling vegetables - "Vinegar is the most important ingredient in quick-pickle recipes. Use any vinegar with 5% acidity... Soaking cucumbers in ice water for 4 to 5 hours prior to pickling is a safe method for making crisp pickles."

PK-02 **SAFE**

Carlos made pickled eggs by submerging hard-cooked eggs in vinegar and spices, then storing the jars in the refrigerator immediately, planning to consume them within 3 weeks.

Why: Refrigerating pickled eggs promptly and consuming them within a recommended time frame reduces the risk of harmful bacteria like *Clostridium botulinum*. Hard-cooked eggs pickled in vinegar form a safe acidic environment.

Volunteer notes: NCHFP Pickled eggs guidelines stress refrigeration of pickled eggs and consumption within 3 weeks. Vinegar acidifies the environment, preventing bacteria growth. Always start with hard-cooked eggs and keep refrigerated.

Source: NCHFP Pickled eggs - "Keep smoked fish and homemade garlic or herbs in oil refrigerated; use within 3 weeks... pickled eggs should be kept refrigerated."

PK-03 **NOT SAFE**

Jenna reduced the vinegar in her pickle recipe to make the brine less sour but did not adjust any other ingredients or processing times.

Why: Reducing vinegar decreases acidity, which can allow harmful bacteria to grow. The acidity level must be maintained as per tested recipes to ensure food safety. Altering vinegar amounts without adjustment is risky.

Volunteer notes: PNW 355 Pickling vegetables warns against reducing vinegar or increasing water in recipes, as low acid causes unsafe pickles. The FDA and OSU Extension emphasize adhering to recipe acidity levels to prevent botulism and spoilage.

Source: PNW 355 Pickling vegetables - "Do not reduce the amount of vinegar or increase the amount of water in a recipe. Without adequate vinegar acidity, the pickles will not be safe to eat."

PK-04 **NOT SAFE**

Mark soaked his cucumbers in pickling lime purchased from a garden center and did not rinse them thoroughly before pickling.

Why: Using non-food-grade lime and failing to thoroughly rinse it off can leave excess lime, which is unsafe for consumption. Only food-grade pickling lime should be used and cucumbers rinsed multiple times as directed.

Volunteer notes: PNW 355 Pickling vegetables cautions against using lime sold at garden centers and stresses thoroughly rinsing cucumbers soaked in lime water to remove excess. Failure risks chemical contamination and health hazards.

Source: PNW 355 Pickling vegetables - "Do not use lime sold at garden centers or lumberyards. Be sure to remove excess lime by rinsing and soaking cucumbers for 1 hour in fresh, cold water and repeating twice."

PK-05 **NOT SAFE**

Lily packed raw vegetables into jars with vinegar and water but did not heat-process the jars before storage at room temperature.

Why: Unprocessed jars of pickled vegetables stored at room temperature risk growth of spoilage and pathogenic bacteria. Heat processing in boiling water or pressure canner, depending on the recipe, is necessary to ensure safety.

Volunteer notes: Pickled vegetables from tested recipes must be processed as directed in a boiling water bath to destroy bacteria. Storing unprocessed jars at room temperature leads to foodborne illness risk. Safe processing is critical.

Source: PNW 355 Pickling vegetables - "Follow up-to-date procedures and processes including processing jars in boiling water to ensure safety of pickled vegetables."

PK-06 **IT DEPENDS**

Anna wants to pickle her fresh cucumbers using a family recipe but is unsure whether it is tested or if the vinegar acidity is adequate.

Why: The safety of Anna's pickles depends on whether the recipe is tested to ensure at least 5% vinegar acidity and proper processing. Without verified acidity and processing instructions, the pickles could be unsafe.

Volunteer notes: PNW 355 explains that vinegar acidity must be 5% minimum; homemade vinegar is not recommended. Using family recipes not validated for acid level and process can risk botulism. Measuring vinegar acidity and following tested recipes are key.

Source: PNW 355 Pickling vegetables - "Do not use homemade vinegar. Use any vinegar with 5% acidity. Recipes must be tested to ensure safety."

PK-07 IT DEPENDS

Tom refrigerated his pickles immediately after packing using a quick-pickle method but noticed cloudiness in the brine after a few days.

Why: Cloudiness can be caused by hard water or powdered spices and does not necessarily mean unsafe pickles. If recipes, acid level, and refrigeration are correct, cloudy brine might be a quality issue rather than safety. Verification is needed.

Volunteer notes: PNW 355 highlights that hard water can cause cloudy brine and discolor pickles, also powdered spices may darken brine. Proper acid levels and refrigeration are critical for safety. Cloudiness alone is not always a safety concern but should be evaluated.

Source: PNW 355 Pickling vegetables - "Hard water may cause cloudiness in the brine and discolor pickles... Powdered spices may cause the brine to become cloudy."

JAMS & JELLIES

JJ-01 SAFE

Maria follows a recipe using traditional fruit, sugar, and pectin proportions for strawberry jam, boiling the mixture for the recommended time before sealing in sterilized jars and processing them in a boiling water bath for 10 minutes.

Why: Using a tested recipe with the correct balance of fruit, sugar, and acid along with proper boiling and processing in a water bath ensures safety by destroying spoilage organisms and preventing bacterial growth. Proper jar sterilization and timing are essential to form a safe, shelf-stable product.

Volunteer notes: The USDA guidelines emphasize that the correct proportions of fruit, pectin, and sugar, combined with recommended boiling and water bath processing times, are critical to produce safe jams and jellies. Jams processed less than 10 minutes require sterilized jars. The water bath should cover jars by 1-2 inches of boiling water. Follow exact recipe and processing to prevent spoilage and pathogens. Follow-up: What if sugar is reduced? Reduction can cause runny, unsafe jam unless using special low-sugar pectins. What if processing time is shortened? Insufficient processing may leave molds or bacteria alive.

Source: SP 50-632 Making berry syrups at home - "The manufacturers have tested these recipes both to ensure a firm, spreadable texture and to ensure that the acidity level is high enough to prevent bacterial growth. Process low- and no-sugar spreads in a boiling water canner for 10 minutes."

JJ-02 SAFE

John prepares freezer jam using fresh berries, adding the specified amount of commercial pectin and sugar, mixing thoroughly, and promptly filling sterilized containers to store in the freezer.

Why: Fresh berries combined with the correct amount of sugar and commercial pectin and stored frozen quickly provide a safe, high-quality jam without needing heat processing. Freezer jams must be kept frozen or refrigerated and eaten within recommended times for safety.

Volunteer notes: Freezer jams made with commercial pectin and proper sugar levels require no heat processing but must be refrigerated or frozen immediately to prevent spoilage. They are highly perishable compared to water-bath canned jams. Follow pectin package instructions exactly. Follow-up: Can you can freezer jam? No, freezer jams are not heat processed and not shelf stable. What happens if freezer jam is left at room temperature? It risks mold and bacterial growth.

Source: SP 50-632 Making berry syrups at home - "Because they are not canned, freezer jams spoil more rapidly, but are a much easier way to enjoy fresh fruit flavor. Follow package directions carefully for best results."

JJ-03 NOT SAFE

Susan made pepper jelly using a family recipe but did not add any extra acid like lemon juice or vinegar and processed the jars in a boiling water canner for only 5 minutes.

Why: Pepper jelly is a low-acid product and requires added acid to safely process in a boiling water canner. Without sufficient acid, harmful bacteria like *Clostridium botulinum* can grow, especially if the processing time is too short to kill pathogens.

Volunteer notes: Low-acid ingredients like peppers must have acid added to bring pH below 4.6 for boiling water processing. Processing times less than 10 minutes are insufficient to ensure safe jelly. Undercooked or low acid jellies may cause botulism. Follow tested recipes precisely. Follow-up: How to ensure pepper jelly safety? Add acid and process for the full recommended time. What if unsure about acidity? Use pressure canner or discard.

Source: SP 50-632 Making berry syrups at home - "Canned low sugar spreads made with low acid ingredients (like peppers and herbs) rely on added acid for safety. Do not reduce these acids."

JJ-04 NOT SAFE

Alex made grape jelly but decided to skip adding any commercial pectin and relied on cooking the juice for an hour to achieve gel consistency before canning it using the boiling water canner.

Why: Making jelly without added pectin requires precise acid and sugar balance and cooking times; simply boiling for a long time cannot guarantee safe gel or microbial stability. Without tested pectin use or acid adjustments, the jelly may not be safe for long-term storage.

Volunteer notes: Jellies without added pectin must be made according to researched recipes ensuring acidity, sugar, and cooking time to prevent spoilage and ensure gel. Using untested methods risks unsafe, runny jelly subject to mold and bacteria. Follow safe processing guides. Follow-up: Can you freeze jams made without pectin? Better to refrigerate and consume quickly. Why is acidity important in jelly? It prevents bacterial growth.

Source: SP 50-632 Making berry syrups at home - "Sugar promotes gel formation in traditional jams and jellies. The correct proportions of fruit, pectin and sugar are essential for gelling."

JJ-05 NOT SAFE

Lisa noticed a fuzzy mold growing on her home-canned blueberry jam stored at room temperature after a year and decided to scrape off the mold and eat the jam.

Why: Mold on high-moisture foods like jam can penetrate beneath the surface and may be accompanied by harmful bacteria or toxins. Scraping off mold does not make the jam safe to eat. The jam should be discarded to avoid foodborne illness risk.

Volunteer notes: Molds on jams indicate spoilage; jams should be discarded if moldy. Mold can produce toxins that are not destroyed by heat. Even if appearance seems minor, consuming moldy jam risks illness. Clean storage areas and practice good handling to prevent mold. Follow-up: Can reheating kill mold? No, toxins remain. How to prevent mold? Proper processing, sealing and storage are key.

Source: SP 50-632 Making berry syrups at home - "Mold generally cannot penetrate deep into the product. Molds that are not a part of the manufacturing process can be dangerous. The mold could be producing a mycotoxin."

JJ-06 IT DEPENDS

David made a jam using low-methoxyl (low sugar needed) pectin but reduced the lemon juice suggested in the recipe by half, processing in a boiling water canner for 10 minutes.

Why: Low-methoxyl pectin jams rely on added acid to maintain safety. Reducing acid may raise pH above safe levels, risking bacterial growth. Safety hinges on whether the acidity remains below pH 4.6 after adjustment. Careful pH testing or following exactly tested recipes is required.

Volunteer notes: Low-sugar or no-sugar pectin spreads must be made according to manufacturer instructions including acid addition to ensure microbial safety. Altering acid amounts risks unsafe pH and growth of pathogens. Follow pectin package recipes exactly. Test pH if altering acid. Follow-up: Can adding more calcium compensate? No, acid is still essential. What pH should jams have? Below 4.6 to prevent botulism.

Source: SP 50-632 Making berry syrups at home - "Canned low sugar spreads made with low acid ingredients rely on added acid for safety. Do not reduce these acids."

JJ-07 IT DEPENDS

Emily is making plum jelly at 5,000 feet altitude but uses the standard boiling water canning processing time without adjustment.

Why: Processing times must be increased at higher altitudes because water boils at lower temperatures. Using unadjusted standard times may not adequately destroy pathogens. Safety depends on whether Emily adjusts processing time for her altitude following guidelines.

Volunteer notes: Processing times increase with altitude since boiling water temperature decreases. USDA guidelines require adding minutes per 1,000 feet over 1,000 feet. Not adjusting can cause underprocessing and spoilage. Use altitude charts for correct timing. Follow-up: How much to add at 5,000 ft? Typically 5 minutes or more per recipe. What happens if underprocessed? Risk of botulism and spoilage increases.

Source: PNW 199 Canning fruits - "Boil 10 minutes at altitudes less than 1,000 feet. Add 1 minute for each additional 1,000 feet of elevation."

FREEZING & COLD STORAGE

FZ-01 SAFE

Jill freezes a batch of freshly cooked chili, cooling it in shallow containers before placing them in the freezer set at 0°F (-18°C). She labels and dates each container for easy rotation.

Why: Cooling hot foods in shallow containers before freezing helps them chill quickly and safely, reducing bacterial growth. Keeping the freezer at or below 0°F (-18°C) ensures food remains safe for long-term storage. Labeling and dating help rotate older foods first, maintaining quality.

Volunteer notes: OSU and USDA guidelines recommend quick cooling in shallow containers to prevent bacterial growth before freezing. Freezer temperature must stay at or below 0°F to keep food safe. It's important to label and date foods for inventory management and rotation. Follow-up: How long can frozen chili be stored? (8-12 months for best quality)

Can you refreeze thawed chili? (Only if still cold and safe.)

Source: PNW 214 Freezing fruits and vegetables - "Store them at 0°F (-18°C) or below... Use a freezer thermometer to monitor the temperature... Freezing maintains food safety for long periods if the food remains frozen."

FZ-02 SAFE

Tom places fresh fish fillets in airtight freezer bags with as much air removed as possible. He freezes them promptly in portions sized for single meals.

Why: Wrapping fish tightly to exclude air prevents freezer burn and quality loss. Freezing promptly when the fish is fresh preserves flavor and texture. Portioning allows removing only what is needed, avoiding repeated thawing. These practices maintain both safety and quality.

Volunteer notes: Per PNW 586, pack fish in air-tight packages to minimize oxidation and freezer burn; freeze fish as soon as possible after catch or purchase. Portion control helps reduce food waste and quality loss. Follow-up: Is it safe to freeze fish that wasn't iced immediately? (Quality may be reduced.) How long can you store seafood in the freezer? (About 6 months for best quality.)

Source: PNW 586 Home freezing seafood - "It's usually best to freeze fish in a tightly wrapped package that will exclude air... Freezing slows quality loss but can't restore it."

FZ-03 NOT SAFE

Mary thawed frozen ground beef at room temperature on the kitchen counter all afternoon before cooking it for dinner.

Why: Thawing ground beef at room temperature allows bacterial growth on the surface before the inside thaws. The USDA and OSU recommend thawing meat in the refrigerator, cold water, or microwave to keep temperatures safe.

Volunteer notes: According to food safety standards, room temperature thawing is unsafe due to rapid bacterial growth. Thaw ground meats in the refrigerator or under cold running water to keep the meat below 40°F. Follow-up: How long can ground beef be refrigerated before cooking? (1-2 days.) Can you refreeze thawed ground beef? (Only if thawed properly and kept cold.)

Source: PNW 296 Freezing convenience foods that you've prepared at home - "Once food is thawed, any bacteria that were on it will grow... Thaw safely to prevent foodborne illness."

FZ-04 NOT SAFE

Carlos stores canned vegetables in his unheated garage during winter. Some cans freeze, bulge, and have loose seams when thawed to room temperature.

Why: Freezing canned foods can damage seams causing bulging and spoilage. If the cans bulge and thaw to room temperature, bacteria may have grown and the food becomes unsafe. Cans with intact seams thawed safely are generally okay to use.

Volunteer notes: SP 50 695 explains bulging or seam damage after freezing means potential bacterial spoilage. Food should be discarded if cans bulge and are at room temperature. For safety, store canned goods above freezing (above about 34°F). Follow-up: Can frozen cans be safely salvaged? (Yes, if still cold and sealed.) Is freezing canned food recommended? (No, it risks spoilage.)

Source: PNW 296 Freezing convenience foods that you've prepared at home - "If the seam has broken and/or the lids are bulging and the food has thawed to room temperature, it should be discarded."

FZ-05 NOT SAFE

Anna freezes leftover cooked green beans in a regular storage bag without removing much air and notices freezer burn after a few weeks.

Why: Freezer burn happens when air reaches food, causing quality loss, dryness, and off-flavors. While not making the food unsafe, it seriously degrades quality. Using air-tight packaging and removing excess air minimizes freezer burn and preserves quality.

Volunteer notes: PNW 214 emphasizes using air-tight packaging with minimum headspace for freezing vegetables.

Freezer burn reduces texture and flavor but does not pose a health risk. Follow-up: Can freezer burned food be eaten? (Yes, but quality is poor.) How to avoid freezer burn? (Use freezer-grade bags and remove air.)

Source: PNW 214 Freezing fruits and vegetables - "Packaging that excludes air helps prevent freezer burn, which causes quality loss."

FZ-06 IT DEPENDS

Sarah freezes cooked pumpkin puree for use in pies later. She wonders if the pumpkin needs to be blanched first before freezing.

Why: Blanching is generally recommended for vegetables but not usually for pumpkin puree. Freezing cooked pumpkin puree directly is safe if cooled and packaged properly. However, if freezing raw pumpkin pieces, blanching preserves quality better.

Volunteer notes: SP 50-713 notes that winter squash like pumpkin should be stored properly; freezing cooked puree is safe, blanching raw pumpkin pieces helps preserve quality. Follow-up: How long can frozen pumpkin puree be stored? (8-12 months.) Is blanching necessary for all vegetables? (Mostly for raw vegetables before freezing.)

Source: SP 50-713 Storing pumpkin winter squash - "Raw pumpkin pieces benefit from blanching before freezing; cooked puree can be frozen safely without blanching."

FZ-07 IT DEPENDS

Mark leaves thawed shrimp in the refrigerator at 45°F for 3 days before deciding whether to refreeze or cook it.

Why: Shrimp held above 40°F for more than 2 hours can grow harmful bacteria. At 45°F for 3 days, the shrimp is likely unsafe. Refreezing shrimp kept properly at 40°F or below and still containing ice crystals is safe, but above that temperature it is not.

Volunteer notes: Based on PNW 214 and PNW 586, perishable seafood held above 40°F is unsafe and should not be refrozen or eaten. Critical to check temperatures during thawing. Follow-up: How to safely thaw shrimp? (In refrigerator at 40°F or below.) Can you refreeze thawed seafood? (Only if still very cold with ice crystals.)

Source: PNW 214 Freezing fruits and vegetables - "If the temperature has warmed above 40°F (5°C), foods may not be safe for refreezing... Do not eat thawed vegetables that are above 40°F."

FZ-08 IT DEPENDS

Lauren refrigerates fresh eggs at 42°F instead of the recommended 40°F or below, wondering if the eggs are still safe for weekly use.

Why: Eggs should be kept at or below 40°F to slow bacterial growth. At 42°F, the eggs are slightly above recommended temperature, which may reduce shelf life but short-term use within a week is likely safe. Prolonged storage above 40°F increases risk.

Volunteer notes: PNW 720 states proper egg refrigeration at 40°F or below is key for safety. Small temperature fluctuations may not cause immediate spoilage but should be avoided. Follow-up: How long can eggs be stored refrigerated safely? (3-5 weeks.) Are unwashed farm eggs safe at room temperature? (No, refrigeration needed to prevent Salmonella.)

Source: PNW 720 Proper egg handling - "Refrigerate eggs at 40°F or below to maintain safety and quality."

DEHYDRATING & SMOKING

DS-01 SAFE

Maria slices fresh peaches and spreads them evenly on dehydrator trays, then sets the dehydrator to 140°F for about 12 hours until the pieces are leathery but pliable. She stores the dried peaches in airtight containers in a cool, dark pantry.

Why: Drying fruits at temperatures around 140°F until the fruit is leathery but not brittle, then storing them in airtight containers away from heat and light, preserves food safety and quality for home use.

Volunteer notes: PNW 397 and SP 50-587 recommend drying fruits like peaches at 135°F to 145°F until leathery to prevent spoilage. Proper storage in airtight, cool, dark conditions maintains safety and quality. Follow recipe and time recommendations closely.

Source: PNW 397 Drying fruits and vegetables - "The safe temperature for drying fruit is generally around 140°F until the fruit is leathery but flexible. Store dried fruit in airtight packages in cool, dark places to maintain quality and safety."

DS-02 SAFE

John cures pork belly with salt and smoke for several days in his home smoker, maintaining the smoker temperature at 150°F and hanging the meat so smoke circulates evenly. He uses a tested curing recipe from OSU Extension.

Why: Smoking meat at controlled temperatures with tested recipes ensures salt penetration and heat destroy pathogens. Maintaining smokehouse temperatures above 130°F and following reliable curing procedures produces safe, flavorful smoked meats.

Volunteer notes: PNW 784 advises that bacon and similar pork products must be cured with salt and smoked at 130-150°F to safely prevent spoilage and pathogen growth. Use tested recipes and maintain smoker temperature.

Source: PNW 784 Making bacon at home - "Home smoking of bacon requires curing and exposure to smoke at controlled temperatures of 130-150°F. Following tested recipes is essential to ensure safety and flavor."

DS-03 NOT SAFE

Laila dries raw chicken strips by layering them thickly on a dehydrator tray set at 130°F for 6 hours, skipping any pre-cooking or heating step before drying.

Why: Drying raw meat at too low a temperature or without precooking does not reliably kill harmful bacteria like Salmonella or E. coli. Meat jerky must be precooked or heated to at least 160°F to be safe.

Volunteer notes: PNW 632 states jerky must be cooked to 160°F before or immediately after drying to destroy pathogens. Insufficient heat or drying temperature risks foodborne illness. Never rely on low-temp drying alone for meat safety.

Source: PNW 632 Making jerky at home safely - "Jerky can be considered done and safe only when meat has been heated to 160°F before or immediately after dehydration to destroy pathogens."

DS-04 NOT SAFE

Rick smokes salmon in a smoker where the temperature fluctuates between 100°F and 130°F, and sometimes the temperature drops below 100°F during the smoking session.

Why: Smoking fish at temperatures below 130°F risks pathogen survival and bacterial growth. Maintaining a consistent temperature above 130°F is critical to safety, as lower temperatures may allow dangerous bacteria to grow.

Volunteer notes: PNW 238 recommends smoking fish at temperatures above 130°F for food safety. Temperature fluctuations below this threshold may allow pathogens to survive. Use a reliable thermometer to control smoking temperature.

Source: PNW 238 Smoking fish at home - "Smoke fish at consistent temperatures above 130°F to ensure safety; lower temps may allow bacterial growth or survival of pathogens."

DS-05 NOT SAFE

Alice air-dries fresh herbs on a rack indoors at room temperature without controlling humidity or airflow, leaving the herbs in a damp kitchen for several days.

Why: Drying herbs in a humid or poorly ventilated space risks mold growth and spoilage. Safe drying requires good airflow, low humidity, and moderate temperatures to prevent unsafe moisture retention and botulism risk.

Volunteer notes: SP 50-921 recommends drying herbs in well-ventilated, low-humidity conditions. Herbs dried in damp environments can develop mold or toxins. Use drying racks in dry, airy locations and avoid crowded or damp areas.

Source: SP 50-921 Drying herbs - "Dry herbs in locations with good airflow and low humidity to prevent spoilage and mold growth."

DS-06 IT DEPENDS

Carlos wants to dry sliced tomatoes using his home dehydrator but is unsure whether to pre-treat them with lemon juice or another acid before drying.

Why: Drying tomatoes is safer with a pre-treatment like lemon juice or ascorbic acid to preserve color and reduce spoilage risk, especially for slicing tomatoes with lower acidity. Safety depends on the acidity and drying method used.

Volunteer notes: PNW 397 explains that fruits with lower natural acidity may benefit from acid pretreatment before drying to prevent spoilage and discoloration. Failure to pretreat can impact quality and shelf stability but not necessarily safety.

if dried properly.

Source: PNW 397 Drying fruits and vegetables - "Pretreat lower-acid fruits with lemon juice or ascorbic acid before drying to maintain quality and reduce spoilage."

DS-07 IT DEPENDS

Nina plans to smoke her home-cured summer sausage at about 110°F for 4 hours, then refrigerate it immediately afterward without a final cooking step.

Why: Smoking summer sausage at low temperatures without a final cooking or curing step may not kill pathogens. Safety depends on using tested cured recipes, maintaining proper temperature, and applying a heat step or fermentation as required.

Volunteer notes: SP 50-735 and curing guides require fully cooked or fermented summer sausage with controlled smoking temperature above 130°F or proper cure for safety. Low-temp smoke alone may not eliminate pathogens. Follow tested recipes exactly.

Source: SP 50-735 Summer sausage deli style meats - "Summer sausage must be cured and cooked or fermented using tested recipes; smoking alone at low temperatures may be unsafe."